

Sharp Solved Families and Constant-Factor Paths in Polynomial Lemniscates

*Root-to-origin segments, critical-value estimates and path containment for Erdős
Problem #1041*

WILL COOK*

7 September 2026

ABSTRACT

Every monic trinomial $z^n + az^m + b$, $1 \leq m < n$, with all roots in the open unit disc has every root-to-origin segment inside $\{|f| < 1\}$, so any two roots are joined through the origin with length less than 2. The proof reduces each radial image to two controlled values. For arbitrary squarefree monic polynomials of degree $n \geq 2$, writing $\mu = \min_{f'(c)=0} |f(c)|$, we also obtain a connector shorter than 2 when $\mu \leq 13/25$, and by scaling a connector shorter than $(5/2)\mu^{1/n}$ in $\{|f| < (25/13)\mu\}$ without a root-location assumption. A separate inverse-map argument treats an isolated simple critical value, while a Poisson identity gives the sharp critical-value mean $\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1)R^{2n/(n-1)}$ for roots in a closed disc of radius R . The solved families, sufficient conditions and obstruction examples do not settle the unrestricted root-connector problem.

1 Monic trinomials, in every degree

Theorem 1.1 (all-degree monic trinomials). *Let $1 \leq m < n$ and*

$$f(z) = z^n + az^m + b,$$

with every zero in the open unit disc. For any zero ζ , the entire segment $[0, \zeta]$ lies in $\{|f| < 1\}$. Distinct zeros ζ_1, ζ_2 are therefore joined by the broken line $\zeta_1 \rightarrow 0 \rightarrow \zeta_2$ of length $\|\zeta_1\| + \|\zeta_2\| < 2$ inside the open unit lemniscate.

Proof. If $\zeta^n + a\zeta^m + b = 0$, then

$$f(t\zeta) = b(1 - t^m) + \zeta^n(t^n - t^m).$$

Vieta gives $|b| < 1$. For $0 \leq t < 1$, the weights $1 - t^m$ and $t^m - t^n$ are nonnegative, so

$$|f(t\zeta)| \leq |b|(1 - t^m) + |\zeta|^n(t^m - t^n) < 1 - t^n \leq 1.$$

At $t = 1$ the value is zero. Concatenating the two segments gives length $|\zeta_1| + |\zeta_2| < 2$. Lean checks [the trinomial radial inequalities and radius budget](#), including the constant-term bound from the root-disc hypothesis. The conversion of those two segment inequalities and their radius-sum bound into the single rectifiable path stated here is an ordinary geometric step. $\square$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

A finite convex certificate. For $f(z) = \sum_{k=0}^n a_k z^k$ and $f(\zeta) = 0$, put $S_j = \sum_{k=0}^j a_k \zeta^k$. Abel summation gives

$$f(t\zeta) = \sum_{j=0}^{n-1} (t^j - t^{j+1}) S_j.$$

For $0 \leq t < 1$, the normalised value $f(t\zeta)/(1 - t^n)$ lies in the convex hull of $S_0, \dots, S_{n-1}$. Bounds on these finitely many values therefore certify a whole segment. In the trinomial case the distinct control values are b and $-\zeta^n$, both strictly inside the unit disc. With more terms, the additional partial sums need their own bounds.

Proposition 1.2 (prescribed-spoke guardrail). *There exist $r \in (0, 1)$ for which every zero of*

$$f_r(z) = z^6 + \frac{1}{5}r^2 z^4 - \frac{1}{5}r^4 z^2 - r^6$$

lies in the open unit disc, yet the radial spoke from the origin to the zero r leaves $\{|f_r| < 1\}$.

Indeed, after $z = rw$, the polynomial factors as $r^6(w^2 - 1)(w^4 + \frac{6}{5}w^2 + 1)$, so all six roots have modulus r . However $f_r(r/2) = -(327/320)r^6$. Any r with $320/327 < r^6 < 1$ proves the assertion. Only this prescribed spoke is excluded; other connectors remain available. The corresponding radial evaluation is checked in [the sextic guardrail](#). The factorisation above explains the root locations.

2 The problem and the general regimes

Problem 2.1 (Erdős #1041). Let $f(z) = \prod_{i=1}^n (z - z_i)$ be monic with $n \geq 2$ and all z_i in the open unit disc. Must two root occurrences be joined by a curve of length less than 2 inside $\{|f| < 1\}$?

Repeated roots give a constant path between two listed occurrences, so the geometric question concerns squarefree polynomials. The original question is Problem 5 of Erdős–Herzog–Pirani [2, Problem 5, p. 139], listed as Problem 1041 in Bloom’s catalogue [1]. Pendyala [4, Theorem 1 and Lemma 1, pp. 1–3] proves the complete degree-four case. His proof separates a close-pair chord case from a radial construction: after normalising a smallest enclosing disc, a four-point angular lemma selects two safe spokes. That vertex need not be a critical point. The results below concern their stated polynomial families and sufficient conditions; no assessment of the current unrestricted problem status or priority is intended.

Theorem 3.1 treats arbitrary geometry when $\mu \leq 13/25$, without a root-location assumption. Its scaling corollary gives length less than $(5/2)\mu^{1/n}$ in the open level $(25/13)\mu$ for every squarefree monic polynomial. The independent 71/10 argument below exhibits how inverse lifts and boundary arcs spend area. The scaled 5/2 consequence is stronger in both length and containment, since $25/13 < 2$; the area-based construction is retained for its proof method and its component-sensitive refinements. The disk-family theorem gives a different sufficient condition, using a simple critical value v with $0 < |v| < 1$ and normalised separation radius $4/3$. The collinear and sparse families retain the target length 2.

Two constructions organise the general estimates. Averaging inverse lifts and boundary arcs converts area into a freely selected connector. Resolving a simple saddle gives a holomorphic inverse whose Dirichlet energy controls its diameter trace. The weighted free-point theorem isolates a complementary critical-value estimate. Section 17 states the remaining selection question and the examples showing which information is insufficient.

3 Low critical values and a uniform path bound

The least critical value is the natural scale for the first merger of root components. A theorem at one fixed critical level therefore supplies a scale-invariant theorem for arbitrary monic polynomials. For a monic degree- n polynomial put

$$K_t = \{z : |f(z)| \leq t\}, \quad \mu = \min_{f'(c)=0} |f(c)|, \quad \rho = \mu^{1/n}.$$

Theorem 3.1 (low-critical parent regime). *Let f be squarefree and monic of degree $n \geq 2$, and let μ be its least critical-value modulus. If $\mu \leq 13/25$, then two distinct roots are joined inside $\{|f| < 1\}$ by a rectifiable curve of length strictly less than 2.*

Corollary 3.2 (scale-free connection). *Every squarefree monic polynomial of degree $n \geq 2$ has two distinct roots joined in $\{|f| < (25/13)\mu\}$ by a curve of length less than*

$$2((25/13)\mu)^{1/n}.$$

In every degree the length can be chosen less than $(5/2)\mu^{1/n}$.

Proof. Put $s = ((25/13)\mu)^{1/n}$ and $g(z) = s^{-n}f(sz)$. Then g is monic and its least critical-value modulus is $13/25$. Apply the theorem and scale the curve back by s . The rescaling step, transporting both the level and the extended variation of an arbitrary curve under $z \mapsto h + cz$, is [the affine transport of a contained rectifiable connector](#); the low-critical input remains ordinary. For $n \geq 3$, $(25/13)^{1/n} \leq (25/13)^{1/3} < 5/4$. For $n = 2$, the segment between the roots has length $2\sqrt{\mu}$ and lies in K_μ . $\square$

The geometric step behind 13/25. Suppose no curve of length less than 2 joins two roots below level 1. Conformal normalisation of an ancestor component gives points $b_1, \dots, b_k$ in the hyperbolic disc with pairwise distance at least D . Write $d_j = d_{\text{hyp}}(0, b_j)$. The inverse-map estimates also give a lower radius $d_* > 0$ and

$$d_j \geq d_*, \quad \sum_j \ell(d_j) \geq x := \log(t/\mu), \quad \ell(d) = -\log \tanh(d/2).$$

The difficulty is to use the pairwise separation without discarding the radial positions of the points. Projecting all the balls onto the circle of directions loses this information: balls at different radii can have overlapping projections. Instead, the open balls of radius $D/2$ around the b_j are intersected with the same hyperbolic circle of radius r . Their intersections are disjoint, possibly empty, arcs. The hyperbolic cosine law gives their angular half-widths

$$w(d, r) = \arccos \left[\frac{\cosh d \cosh r - \cosh(D/2)}{\sinh d \sinh r} \right]_{-1}^1, \quad \sum_j w(d_j, r) \leq \pi,$$

where brackets denote truncation to $[-1, 1]$. Choose radii r_i and weights $\sigma_i \geq 0$. If

$$\ell(d) \leq U + \sum_i \sigma_i w(d, r_i) \quad (d \geq d_*), \quad U > 0,$$

then summing over the roots proves the finite dual bound

$$k \geq \frac{x - \pi \sum_i \sigma_i}{U}.$$

The summation and supremum step from the majorant and the slice bounds to the boxed inequality is [the finite dual arity floor, with the packing bound and the boundedness of the dual values as premises](#); the hyperbolic supplier of those premises is ordinary. The nonnegative weights combine valid bounds from several radii; the summation is elementary once the uniform majorant has been proved. Establishing that majorant and the geometric separation estimates is the substantive input. The resulting lower bound for k enters the length–area comparison until a short boundary connection is forced. The complete comparison proof and rational certificate are in [the angular budget low critical closure](#). The certified terminal time is $635762889599/10^{12}$, and $(13/25) \exp(635762889599/10^{12}) < 1$.

An independent area proof. The next theorem gives a weaker numerical bound through an argument that needs no circle-slice certificate. It makes the area allocation explicit.

Theorem 3.3 (unconditional constant-factor path). *For every monic polynomial f of degree $n \geq 2$, two zero occurrences are joined by a possibly degenerate path of length at most*

$$\frac{71}{10} \rho$$

inside $K_{2\mu}$. If f is squarefree, their locations are distinct. If $\mu \leq 1/2$, the construction may be chosen inside $\{|f| < 1\}$ with length at most 5.7.

The constant is the rationally certified specialization of a two-parameter bound. If the selected working component contains $k \geq 2$ roots, then for every $r \in (0, 1)$ and $\lambda > 1$ the proof constructs a path in $K_{\lambda\mu}$ whose length is at most

$$\sqrt{\frac{2}{k}} \left(\frac{\sqrt{2}r}{(1-r)^2} + \lambda^{1/n} \left(\sqrt{\log(\lambda/r)} + \frac{\pi}{\sqrt{\log \lambda}} \right) \right) \rho. \quad (\text{CF})$$

For $n \geq 3$, the choice $\lambda = 2$, $r = 3/20$ makes the bracket less than $71/10$; degree two has the exact root-segment bound 2ρ .

Proof. A repeated zero gives a constant path of length zero, so assume f is squarefree; then $\mu > 0$. Write $K_t = \{|f| \leq t\}$, C_T for the component of K_T containing a minimising critical point, and $A(\sigma) = \text{Area}(K_\sigma \cap C_T)$. Degree two is the root segment of length 2ρ inside K_μ ; assume $n \geq 3$.

If a regular component C' of $K_\sigma \cap C_T$ holds k' roots, then $|dz| = \sigma d\varphi/|f'|$ on $\partial C'$ and $\int d\varphi = 2\pi k'$. Cauchy–Schwarz and coarea give

$$\mathcal{H}^1(\partial C')^2 \leq 2\pi k' \sigma A'(\sigma). \quad (\text{CF1})$$

Pólya's inequality $A(T) \leq \pi T^{2/n}$ [7, Theorem 1] and the change of measure $d\sigma/\sigma$ on $[\mu, T]$, $T = \lambda\mu$, produce a non-critical level $t \in [\mu, T]$ with $tA'(t) \leq \pi T^{2/n}/\log \lambda$. The component $C_t \subseteq C_T$ still holds $k \geq 2$ roots, and

$$\mathcal{H}^1(\partial C_t) \leq \pi \sqrt{2k/\log \lambda} T^{1/n}.$$

The level t is chosen from the mean of $\sigma A'(\sigma)$ *before* any path is assembled. That is the only device that removes the logarithmic divergence of A' at a critical level.

For arguments avoiding the finitely many critical-value rays, each root in C_t lifts along a value-ray to ∂C_t . Split every lift at level $r\mu$. On a one-root lobe the inverse of f/μ is univalent, so Koebe distortion bounds the low pieces pointwise; the precritical identity

$$\sum_i \frac{\mu^2}{|f'(z_i)|^2} \leq \rho^2 \quad (\text{CF2})$$

then aggregates them by Cauchy–Schwarz, paying $\sqrt{2/k}$ rather than once per root. Coarea and (CF1) bound the high pieces only after averaging in the argument. Choose one argument realising that average, order the lift endpoints cyclically on ∂C_t , and hop along the shortest adjacent boundary arc. Averaging the two lifts and the k arcs over adjacent pairs gives (CF). Explicitly, the three contributions to that pair average satisfy

$$\begin{aligned} \frac{2}{k} \sum_i \ell_i^{\text{low}} &\leq \frac{2r\rho}{\sqrt{k}(1-r)^2}, \\ \frac{2}{k} \sum_i \ell_i^{\text{high}} &\leq \sqrt{\frac{2}{k}} \sqrt{\log(\lambda/r)} T^{1/n}, \\ \frac{\mathcal{H}^1(\partial C_t)}{k} &\leq \sqrt{\frac{2}{k}} \frac{\pi T^{1/n}}{\sqrt{\log \lambda}}. \end{aligned}$$

The low estimate is pointwise in the ray; the high estimate holds for the single ray chosen from its mean. The boundary estimate uses the level already selected. Thus these estimates apply together, and $T^{1/n} = \lambda^{1/n}\rho$ gives the three terms in (CF). For the final assertion, the repeated-zero case is already a constant path, and in degree two the root segment has length $2\rho = 2\sqrt{\mu} \leq \sqrt{2}$. For $n \geq 3$, take $\lambda = 2$ and $r = 3/20$. Since $k \geq 2$ and $\mu \leq 1/2$, (CF) gives

$$\begin{aligned} \text{length} &\leq \frac{60\sqrt{2}}{289} \mu^{1/n} + (2\mu)^{1/n} \left(\sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}} \right) \\ &\leq \frac{60\sqrt{2}}{289} + \sqrt{\log(40/3)} + \frac{\pi}{\sqrt{\log 2}} \\ &= 5.676476 \dots < 5.7. \end{aligned}$$

Here $2\mu \leq 1$, and the mean-value choice may be taken at a regular level strictly below the top of its positive-measure window. Thus the entire path lies in $\{|f| < 1\}$. $\square$

The component-arity, capacity and two-root persistence-window refinements are retained with their complete hypotheses in [unconditional constant factor bound](#) and [minimal hub window join](#). Their parent conclusions under $\mu \leq 1/2$ are already covered by Theorem 3.1; their additional content is control of a specified component or a specified descent arc. The exact distinction between a simple minimum and a unique minimum is preserved with the persistence-window argument.

4 Critical proximity and straight-path obstructions

A critical point gives a different way to select two nearby roots, without assuming a sparse coefficient pattern. The logarithmic-derivative balance prevents one root from being arbitrarily closer to the critical point than all the others. This yields a bound on the sum of two distances. To turn that bound into a path theorem, both connecting segments would also have to stay inside the lemniscate; the examples below show why that extra step cannot be assumed.

Theorem 4.1 (critical geometric-mean proximity). *Let $n \geq 2$, let $z_1, \dots, z_n, c \in \mathbb{C}$ with $c \neq z_k$ for every k , and suppose*

$$\sum_{k=1}^n \frac{1}{c - z_k} = 0.$$

If $r > 0$ is determined by

$$r^n = \prod_{k=1}^n |c - z_k|,$$

then there are distinct indices i, j such that

$$|c - z_i| + |c - z_j| \leq 2r.$$

The proof chooses the two smallest distances $d_i \leq d_j$. The reciprocal balance gives $d_j \leq (n-1)d_i$, while minimality and the geometric-mean identity give $d_i d_j^{n-1} \leq r^n$. The role of the balance is visible on setting $t = d_j/d_i$: it restricts t to $[1, n-1]$, where

$$1 + t \leq 2t^{(n-1)/n}.$$

Indeed, $\log(2t^{(n-1)/n}/(1+t))$ vanishes at $t = 1$ and has derivative $(n-1-t)/(nt(1+t)) \geq 0$ on that interval. Consequently $d_i + d_j \leq 2(d_i d_j^{n-1})^{1/n} \leq 2r$. Without the reciprocal balance, the required restriction on t would be missing. The complete complex theorem, including the selector and both inequalities, is checked as [the critical geometric-mean proximity](#).

Corollary 4.2 (two nearest roots). *If the roots lie in the open unit disc and c is a non-root critical point, the two nearest roots to c have total distance strictly below 2.*

Proof. Apply the closed-disc two-nearest-root inequality after scaling by any radius $R < 1$ containing all the roots. The ordinary reduction is recorded in the [global two-nearest-root budget](#); Lean checks its algebraic core as `two\ add\ le\ two\ of\ disk\ inverse\ balance`. $\square$

This is not silently promoted to a path theorem. The same source module checks two exact barriers to the most tempting completions.

Proposition 4.3 (two straight-path no-go families). *There is a monic quintic with all roots in the open unit disc and a non-root critical point c whose unique nearest root has a point on the straight spoke to c outside $\{|f| < 1\}$. There is also a monic cubic with all roots in the open unit disc such that the midpoint of every pair of distinct roots lies outside $\{|f| < 1\}$.*

The certificates are exact rational-algebraic computations, checked as the [unique nearest-spoke escape](#) and the [cubic all-straight midpoint obstruction](#), and the displayed proposition itself, with both quintic and cubic exhibited as polynomials and the nearest-root condition quantified over every zero, records those same two evaluations. Lean checks the proposition as [the complete straight-path obstructions](#). The first defeats a tie-breaking escape hatch; the second defeats every straight root-pair segment. Together with Theorem 4.1, they isolate the real gap: critical-point proximity selects a short metric pair, but universal containment requires curved or topological geometry.

5 Solved polynomial families

The following families cross that containment gap by controlling entire spokes or segments. The translated cubic fibre theorem has a complete rectifiable-path formalisation. For the other two families, the displayed proofs identify the checked kernels and the ordinary geometric steps.

5.1 Translated cubic quotient fibres

Theorem 5.1 (translated cubic quotient fibres). *Let $q \geq 2$, $h \in \mathbb{C}$, and let P be a monic cubic. Put*

$$f(z) = P((z - h)^q).$$

If all zeros of f lie in the open unit disc and f has at least two distinct zeros, then two distinct zeros of f are joined by a two-segment path of length strictly less than 2 contained in $\{|f| < 1\}$.

For cubic roots r, s, v , the real charges $\Re(\overline{rs} + v)$ sum to more than -3 , so one is greater than -1 . For that root, AM–GM and the exact cyclotomic cancellation $(1 - t)(1 + t + t^2) = 1 - t^3$ prove that its complete origin spoke is safe. Lean checks [the safe cubic root-spoke fan-in](#). Pulling the spoke back under $y \mapsto y^q$ gives q safe spokes from h ; two different fibre points are joined through h with length $2|y| < 2$. The finite root-of-unity mean-square argument and fibre assembly are also formalised in [the complete translated cubic fibre theorem](#). Its conclusion constructs the explicit continuous rectifiable two-segment path with strict containment and variation below 2. The exact source files occur in the successful compiled dependency closure of [the source-bound Lean build and axiom-audit record](#).

5.2 Primitive sparse quintics

Theorem 5.2 (primitive sparse quintics). *Let*

$$p(z) = z^5 + az^4 + bz + c$$

be monic with all five root occurrences in the open unit disc. Then two distinct root occurrences w_i, w_j satisfy

$$|bw_i + c| < 1, \quad |bw_j + c| < 1,$$

and the corresponding two radial spokes join them through the origin inside $\{|p| < 1\}$ with total length strictly below 2. If the two occurrences coincide, the associated root-to-root path is degenerate.

The stronger closed-disc selector has non-strict tails; if $a \neq 0$ the selected tails are strict, while for $a = 0$ strictness is exactly the interior-root condition. The phase-sensitive boundary theorem uses the first three Newton moments and a cubic separator. Its harmonic extension controls mixed interior/boundary configurations and shows that an unsafe interior tail contributes at most $2/31$. Lean checks the finite selectors [two boundary tail energies below one](#) and [two interior tail energies below one](#). The attached source snapshot also contains [a complete primitive-quintic path declaration](#). This revision admitted that module to the checked ErdosProblems root; AxiomAudit prints `complete_primitive_quintic` against the standard axioms. The finite checked kernels and the full geometric theorem remain distinct layers: the former were already in the root, and the latter is now a checked implementation rather than an unchecked source file.

5.3 Sharp collinear roots

The complete all-degree collinear theorem appears in Section 14. It gives the sharp Chebyshev constant, not merely an existence bound, and therefore belongs to the solved-family layer rather than the frontier layer. Lean checks [the constrained monic-comparison alternation](#) and [its Chebyshev comparison-bound consumer](#); affine normalisation, root-gap selection, and transport back to the original line are ordinary.

6 Critical-value separation in every degree

The solved families of Section 5 are unconditional and complete, and each is confined to a shape of root configuration. This section gives a hypothesis that is not a shape at all. It constrains only the critical spectrum, it holds in every degree above an exact cutoff, and where it holds it gives the target constant 2 rather than a constant multiple of it.

Theorem 6.1 (disk-family critical-value separation). *Let f be monic of degree $n \geq 3$, let c be a simple critical point, and put $v = f(c) \neq 0$. Fix $w_0 \in [0, 1]$ and $S > \max(w_0, 1 - w_0)$. Suppose every other critical point d satisfies*

$$\left| \frac{f(d)}{v} - w_0 \right| \geq S.$$

Put $p = w_0(1 - w_0)$. Then two distinct roots are joined inside $\{|f| \leq |v|\}$ by a curve Γ satisfying

$$\text{length}(\Gamma)^2 \leq 2|v|^{2/n} \left(\frac{S}{n-1} \right)^{2/n} \log \frac{S^2 + S + p}{S^2 - S + p}. \quad (1)$$

Corollary 6.2 (uniform radius $4/3$). *Let f be monic of degree $n \geq 3$ with all roots in the open unit disc. If c is a simple critical point with $0 < |f(c)| < 1$ and, for some $w_0 \in [0, 1]$, every other critical point d satisfies*

$$\left| \frac{f(d)}{f(c)} - w_0 \right| \geq \frac{4}{3},$$

then two roots are joined inside $\{|f| < 1\}$ by a curve of length strictly below 2. In degree three the branch-centred choice $w_0 = 1$ already works with $4/3$ replaced by $6/5$.

Corollary 6.3 (a parent-problem regime). *Let f be monic of degree $n \geq 3$ with all roots in the open unit disc, and let c be a simple critical point with $v = f(c)$ and $0 < |v| < 1$. If some real centre $w_0 \in [0, 1]$ admits a radius $S \geq 4/3$ such that $|f(d)/v - w_0| \geq S$ for every other critical point d , then Erdős Problem #1041 holds for f .*

The hypothesis $|v| < 1$ is required: normalised separation controls a connector in $\{|f| \leq |v|\}$, which lies in the unit lemniscate only when $|v| < 1$. The Fekete chain supplies $|v| < 1$ at a minimum-modulus critical point, but an arbitrary selected critical value need not. The hypothesis is sufficient, not necessary: it does not assert that some critical value always admits such a disk. The cubic $z^3 + (3/100)z - 3/4$ shows in particular that the older separation-two condition does not cover all polynomials.

Proof of Theorem 6.1. Set $\rho = |v|^{1/n}$ and

$$P(w) = \frac{f(c + \rho w)}{v}.$$

Then $P(0) = 1$, $P'(0) = 0$, $P''(0) \neq 0$, and its leading coefficient is $|v|/v$, of modulus one. This is the normalization used below; lengths scale back by ρ .

Let $Q = D(w_0, S)$ and let U be the component of $P^{-1}(Q)$ containing 0. The hypotheses put both 0, 1 and the segment $[0, 1]$ in Q , while 0 is the only critical point in U . The restriction $P : U \rightarrow Q$ is proper, and U is simply connected by the maximum principle. The component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1], applied to $(P - w_0)/S$, gives $\deg(P|_U) = 2$ from the one simple ramification point. Thus $1 - P$ has only its double zero at 0 on U and admits a single-valued square root $\xi^2 = 1 - P$. This map is proper and locally injective onto

$$\tilde{Q} = \{\xi : |\xi^2 - (1 - w_0)| < S\},$$

the connected simply connected double cover of Q . It is therefore a conformal bijection. Its inverse Z is the global resolved branch, and $Z([-1, 1])$ joins the two distinct points over $P = 0$ inside $\{|P| \leq 1\}$. This component argument is what prevents the local inverse germ from changing sheets or escaping during continuation.

Put $a = 1 - w_0$. The conformal map

$$\zeta(\xi) = \xi \sqrt{\frac{S}{S^2 + a\xi^2 - a^2}} : \tilde{Q} \rightarrow \mathbb{D}$$

sends $[-1, 1]$ to $[-q, q]$, where $q^2 = S/(S^2 + p)$.

The Bergman segment inequality is the following estimate. Let $\Phi : \mathbb{D} \rightarrow U$ be conformal with $\text{Area}(U) < \infty$, and let $0 < q < 1$. The monomials $\sqrt{(m+1)/\pi} z^m$ are orthonormal for area measure on $\mathbb{D}$, so the Bergman space $A^2(\mathbb{D})$ has reproducing kernel $K(z, w) = \pi^{-1}(1 - z\bar{w})^{-2}$, and $\|\Phi'\|_{A^2}^2 = \text{Area}(U)$. Choose a measurable unimodular θ with $\theta\Phi' = |\Phi'|$ on $[-q, q]$. The reproducing property, Cauchy–Schwarz and the positivity of K on $[-q, q]^2$ give

$$\left(\int_{-q}^q |\Phi'(x)| dx\right)^2 \leq \text{Area}(U) \int_{-q}^q \int_{-q}^q \frac{dx dy}{\pi(1 - xy)^2} = \frac{2}{\pi} \log \frac{1+q^2}{1-q^2} \text{Area}(U).$$

Here $(1+q^2)/(1-q^2) = (S^2 + S + p)/(S^2 - S + p)$. Applying this estimate to $\Phi = Z \circ \zeta^{-1}$ gives

$$\text{length}(Z([-1, 1]))^2 \leq \frac{2}{\pi} \log \frac{S^2 + S + p}{S^2 - S + p} \text{Area}(U). \quad (*)$$

The area estimate concerns this particular two-sheeted component. For $1 - w_0 < S' < S$, let U' be the component of $\{|P - w_0| < S'\}$ containing 0. It is a proper regular component containing exactly two zeros of $P - w_0$, counted with multiplicity. The exterior-fibre capacity gap gives

$$\frac{\text{cap}(\overline{U'})^n}{S'} < \frac{1}{n-1}.$$

Pólya’s area–capacity inequality $\text{Area}(K) \leq \pi \text{cap}(K)^2$ [15, printed pp. 280–282], in the form [7, Theorem 6], then gives $\text{Area}(U') < \pi(S'/(n-1))^{2/n}$. The components U' exhaust U as $S' \nearrow S$, so $\text{Area}(U) \leq \pi(S/(n-1))^{2/n}$. Substitute this in $(*)$ and scale lengths by ρ to obtain (1). Dubinin’s relative area inequality [14, Theorem 1] assumes a full n -fold covering of an annulus; the proof above uses the absolute inequality only. $\square$

For $4/3 \leq S \leq 2$ and $p \geq 0$,

$$\frac{S^2 + S + p}{S^2 - S + p} \leq \frac{S+1}{S-1} \leq 7, \quad \frac{S}{n-1} \leq 1,$$

and $\log 7 < 2$. Thus (1) is strictly below $4|v|^{2/n} < 4$ under the corollary’s hypotheses. The cubic constant follows from $(3/5)^{2/3} \log 11 < 2$.

Boundary. The two-sheeted component, its square-root uniformisation, the Bergman segment inequality, the exterior-fibre capacity gap, Pólya’s inequality and the exhaustion are ordinary mathematics. Their complete immutable source is the [disk-family critical-value separation record](#). The separate [Lean numerical kernel](#) checks the coefficient bound for $n \geq 3$, $4/3 \leq S \leq 2$, the cubic $S = 6/5$ inequality, and the implication from the squared bound to length below 2. It does not formalise the analytic hypotheses producing (1); the adjacent axiom-audit source audits those numerical declarations rather than the ordinary analytic theorem. Nothing in this section asserts a separated disk always exists, covers the near-tie or multiple-saddle strata, or proves the unrestricted problem.

Pendyala’s degree-four theorem [4, Thm. 1] has no critical-spectrum hypothesis. The present separation criterion applies in every degree $n \geq 3$ and supplies an explicit inverse-map length estimate.

7 Three exact covering obstructions

None of the following is a counterexample to Erdős #1041. Each kills a covering reading of a sufficient regime. Rational cores are checked in [revision r2 exact cores](#); the ordinary arguments are in [exact obstructions r2](#).

Proposition 7.1 (no separation-or covering). *Let $f(z) = z^3 + (3/100)z - 3/4$. Every root lies in the open unit disc, both critical points are simple, the critical values lie on distinct positive rays, $\mu > 13/25$, and*

$$|1 - f(c_-)/f(c_+)| < 2/375 < 2.$$

The whole proposition, including the least-modulus assertion rather than a supplied minimum, is checked in Lean as [the complete separation counterexample](#); the distinct quartic file SeparationCounterexampleR13.lean is not this cubic example.

Proposition 7.2 (one-root gamma bound is false). *Let $p(z) = z^8 - (3/2)z$ and let C be the component of $\{|p| \leq 1\}$ containing the origin. Then C contains exactly one zero and a neighbourhood of the closed disc of radius $5/8$, so $\mathcal{H}^1(\partial C) > 5\pi/4$. The constant $\Gamma(1/4)^2/(2\sqrt{\pi})$ is at most $(\pi/2)(1 + \sqrt{2}) < 5\pi/4$.*

An arbitrary one-root component has no degree-uniform perimeter bound: the Runge-approximation construction in the accompanying research addendum [21, Proposition 2] produces monic level-one examples with arbitrarily long analytic boundaries. That construction does not require all roots in the unit disc. The narrower question relevant here requires *every* inverse branch to be unramified at the chosen level: $0 < \sigma < \mu(f)$. For a branch $f(\phi(w)) = \sigma w$, it asks whether $\int_0^{2\pi} |\phi'(e^{it})| dt \leq \beta \sigma^{1/n}$ with an absolute β . Necessarily $\beta \geq 2\pi$, as shown by $z^N - az$ with fixed $a > 1$ and $N \rightarrow \infty$, followed by $a \downarrow 1$. The family $z^N - z$ at level 1 does not supply the required isolated component. Neither general path theorem uses the rejected arbitrary-component bound.

Proposition 7.3 (arity does not force a capacity gap). *Let $g(z) = z^3 - (3/400)z - 3/32$. All roots lie in the open unit disc, $\mu = 187/2000 \leq 1/2$, the first-merge arity is $k_0 = 2$, and the ancestor component at level 2μ has normalised capacity 1.*

8 The Newton value equation

Away from the critical set, define the complex Newton field

$$N(z) = -\frac{f(z)}{f'(z)}.$$

Let $I \subseteq \mathbb{R}$ be an interval on which a differentiable curve z satisfies $f'(z(t)) \neq 0$ and $z'(t) = N(z(t))$. Put $w(t) = f(z(t))$. Kozen and Stefánsson record the following identity as a lemma of Shub, Tischler and Williams [6, Lemma 2.1].

Theorem 8.1 (value equation). *For a polynomial f and a differentiable curve $z : I \rightarrow \mathbb{C}$ on an interval I , assume $f'(z(t)) \neq 0$ and $z'(t) = -f(z(t))/f'(z(t))$ throughout I . Then $w = f \circ z$ satisfies $w' = -w$, and*

$$f(z(t)) = e^{-(t-t_0)} f(z(t_0)) \quad (t, t_0 \in I).$$

The local computation is one line: $w' = f'(z)z' = -f(z) = -w$. Multiplication by e^t and integration on I give the displayed identity. Lean checks the local complex-parameter chain rule as [the Newton-flow value equation](#) and its scaled zero-derivative form as [vanishing of the exponentially scaled value](#). The integrated real-time assertion above is an ordinary consequence.

For an existing trajectory with nonzero initial value, the value moves inward on one positive ray. A zero value remains zero and has no argument. Thus a trajectory starting in $\{|f| < 1\}$ stays there at later times in its interval of existence. The equation neither constructs a global trajectory nor identifies a whole ray preimage with one flow line.

Corollary 8.2 (ray separation). *Let $a < b$, let $z : [a, b] \rightarrow \mathbb{C}$ be continuous, and suppose z is differentiable on (a, b) with $f'(z(t)) \neq 0$ and $z'(t) = -f(z(t))/f'(z(t))$ there. Then*

$$f(z(b)) = e^{a-b} f(z(a)).$$

If the endpoint values are nonzero, they lie on the same positive ray. In particular, critical points whose values lie on distinct positive rays cannot be the endpoints of such a finite connection.

Apply the value equation inside (a, b) and pass to the endpoints by continuity. This does not evaluate $-f/f'$ at a critical endpoint. Lean checks the algebraic contradiction from an assumed exponential endpoint relation in [the distinct-ray endpoint implication](#). Kozen and Stefánsson draw the same consequence for the Newtonian graph: under f , every edge maps onto a segment of a ray through the origin whose endpoints are 0 or critical values [6, §2]. The endpoint behaviour of maximal trajectories is classified there as well [6, Lemma 2.2].

9 Arguments, not moduli

It is tempting to arrange a generic perturbation so that the critical values are pairwise distinct, or that their moduli are pairwise distinct, and to conclude that saddle connections are excluded. Neither is enough.

Two distinct critical values can lie on one ray, and two critical values with distinct moduli certainly can: the ray records the argument, and the modulus is exactly the coordinate the flow contracts. By Corollary 8.2 the invariant that excludes connections is the argument. What a perturbation must therefore achieve is pairwise distinct critical-value *arguments*, which is a condition on $n - 1$ points modulo the circle rather than on their positions in the plane.

The cost of that condition is also checked, and it is small.

Theorem 9.1 (ray-collision locus). *Let $a \neq b$ be complex. Every common translation β for which $a + \beta$ and $b + \beta$ lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \quad r \in \mathbb{R}_{>0}, r \neq 1.$$

So each pair of critical values contributes a one-real-parameter forbidden locus in the translation plane, given in closed form ([the ray-collision parameterization](#)). A finite union of such loci has empty interior, so [an arbitrarily small translation separates the](#)

arguments. Under an explicit margin, a small constant perturbation keeps the roots in the unit disc. Fixed-degree lower semicontinuity in Section 17 passes a dense-generic closed-level length bound to the whole coefficient class. Quantitative forward transport of a selected curve requires additional control, but is not a premise of that existence argument.

10 Local coordinates for perturbative competitors

For $f(z) = \prod_j (z - a_j)$ with $a_j \neq 0$, put $r = \min_j |a_j|$ and $p_m = \sum_j a_j^{-m}$. On $|z| < r$, the exact expansion

$$\log |f(z)| = \log |f(0)| - \Re \sum_{m \geq 1} \frac{p_m}{m} z^m$$

keeps the nonlinear terms as coefficient coordinates. If $q = |z|/r < 1$, the tail after N has modulus at most $nq^{N+1}/((N+1)(1-q))$; differentiated geometric series give the corresponding first and second derivative bounds. The complete reciprocal recurrences, zero-margin staple construction, and quadratic contact test are proved in the long record and reciprocal newton expansion. Their use requires a compatible local contact; the expansion alone supplies no global root-pair selection.

11 A proof gap in the unrestricted argument

The March manuscript's Proposition 12 claims the following load-bearing statement. For $u = -\log |f|$, a connected component V of $\{u > c\}$ carrying $m \geq 2$ simple zeros, and the stated regularity and Morse hypotheses, it constructs, for every $\varepsilon > 0$, an embedded spanning tree $G_\varepsilon \subset V$ with

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^\infty P_V(t) dt + \varepsilon. \quad (5.1)$$

The final theorem uses this estimate, so the issue below cannot be bypassed by calling the proposition auxiliary. Proposition 7 of the unattributed local revision repeats the same estimate.

The estimate itself is false. Take the Cassini polynomial $f_a(z) = z^2 - a^2$ at $a = 9/10$. Its component contains the two roots at distance $9/5$ and satisfies the relevant Morse and critical-value hypotheses. Direct level-length calculation gives the upper majorant

$$4(\sqrt{a^2 + a} - a) < \frac{41}{25}$$

for the right side of (5.1) before ε , whereas every connected set containing both roots has length at least $9/5$. Since $9/5 - 41/25 = 4/25$, choosing a smaller positive ε contradicts the assertion. Thus neither a different local saddle neighbourhood nor a perfect topological decomposition can recover the printed coefficient $1/(2\pi)$. The exact inequality and its abstract tree-budget contradiction are checked in `ErdosProblems.Erdos1041.CassiniTreeBudget`; the level-length majorant is an explicit analytic input, not a kernel-checked integral evaluation.

At an interior index-one critical point p , the proof invokes a Morse chart

$$u = \mu + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because $p \in V$ and V is open, a sufficiently small closed disc around p lies entirely in V . In that disc the full Morse chart has four sectors: two components of $u > \mu$ and two components of $u < \mu$. The level set $\{u = \mu\}$ is a lemniscate graph with a vertex of degree four at p [13, Definition 1.2 and Proposition 2.4]. A global component argument cannot delete one local sector from a disc already contained in V .

This independently diagnoses a proof step, but the Cassini witness above also refutes the proposition's statement. A different route might cut an adjoining regular annulus along a separatrix or regular flow arc before forming the block, retain a four-pronged saddle neighbourhood and change the assembly, or replace the local construction by the ray-cut decomposition proposed below. But no repair can retain (5.1); it must pay a positive attachment cost, select only one short pair instead of spanning every root, or use a different global metric inequality. The shorter descriptions of the same three-ended block do not repair the four-sector topology.

Corollary 8.2 supplies one independent input for a different route: distinct critical-value arguments exclude saddle-to-saddle Newton connections. It does not itself prove the compact planar decomposition, classify all orbit endpoints or provide the metric gluing estimate. Those are separate problems below.

12 Computational searches

Raster searches over sampled root configurations supplied candidate paths through degree ten. They supply no degree-wise upper bound. A raster path becomes a continuous certificate only after every connecting segment and its length have been enclosed. The sample counts, grid distances, and search parameters are retained in the long record. Further searches should track named obstructions: clustered critical values, shrinking attachment windows, component mergers, and distance to the unit level.

13 Exact symmetric benchmark: complementary chords for every binomial degree

There is a complete all-degree model family which isolates the metric geometry without pretending to select paths for arbitrary polynomials. Let $f(z) = z^n - a$, where $n \geq 2$ and $0 < |a| < 1$. After rotation write $a = r^n > 0$, put $c = \cos(\pi/n)$, and set

$$r_* = (1 + c^n)^{-1/n}, \quad \varepsilon = (1 - r^n)^{1/n}.$$

Theorem 13.1 (complementary binomial chords). *Two adjacent zeros of $z^n - a$ can be joined by an explicit polygonal path inside $\{|z^n - a| < 1\}$ of length strictly below 2. For $r < r_*$ the adjacent-root chord itself works. For $r \geq r_*$, two radial legs and an inner adjacent crossing chord work after an arbitrarily small radial contraction. These two constructions meet at $r = r_*$, where*

the outer chord attains $|f| = 1$ at its midpoint and therefore lies only in the closed lemniscate. Open containment at and above the switch uses the inner chord after a radial contraction.

The central chord calculation is exact. On the chord between adjacent roots,

$$\max |z^n - r^n| = r^n(1 + c^n),$$

with the maximum at the midpoint; hence the first construction has precisely the threshold r_* . For $r \geq r_*$, which forces $n \geq 3$ because $r_* = 1$ when $n = 2$, put $t = \varepsilon/c$. The path from r to t to $t\omega$ to $r\omega$ has its crossing chord in the closed lemniscate, and its length is

$$2r - 2\varepsilon \tan\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) < 2r < 2.$$

The radius t is maximal: a larger crossing chord already escapes at its midpoint. Contracting that chord to radius λt with $0 < \lambda < 1$ makes the containment strict while keeping the two radial connections and the strict length bound. For $n = 2$, the adjacent-root diameter is already a strict path of length $2r < 2$.

The proof uses the sharp inequality

$$1 + \cos(n\theta) \leq 2 \cos^n \theta \quad (|\theta| \leq \pi/n),$$

followed by elementary chord geometry and the disk inequality that justifies the contraction. It is an authored ordinary all-degree proof with bounded numerical replay, not a Lean theorem or a priority claim. Its role is a symmetric benchmark: the exact transition says why a direct outer chord fails and why the complementary inner chord repairs it. It gives no path optimality, no theorem for arbitrary monic polynomials, and no solution of Erdős Problem 2.1.

14 The sharp collinear theorem: order buys an exact Chebyshev extremal

Collinearity supplies the global ordering that arbitrary complex root sets lack, and it yields a sharp all-degree solved subcase. Put

$$C_n = \frac{1}{2^{n-1} \cos^n(\pi/(2n))}.$$

Theorem 14.1 (sharp collinear root-diameter bound). *Let f be monic of degree $n \geq 2$, with collinear zero occurrences of diameter D . Then some two zero occurrences are joined by their straight segment, of length at most D , on which*

$$|f| \leq C_n \left(\frac{D}{2}\right)^n.$$

The constant is sharp: equality is attained by the affine images of the scaled Chebyshev root configuration with extreme roots at distance D .

Proof. Repeated zeros give the constant path, so suppose the zeros are distinct. Translation, rotation, and scaling by $R = D/2$ reduce to a monic real-rooted polynomial q with roots

$$-1 = y_1 < \dots < y_n = 1, \quad |f| = R^n |q| \text{ on the root line.}$$

Set $r_n = \cos(\pi/(2n))$ and compare q with the monic polynomial

$$q_*(x) = \frac{T_n(r_n x)}{2^{n-1} r_n^n}.$$

It vanishes at ± 1 and has $|q_*| \leq C_n$ on $[-1, 1]$. Choose a point c_i of maximum $|q|$ in each adjacent gap. If every gap maximum exceeded C_n , then $h = q - q_*$ would have the alternating signs of $q(c_i)$ at the c_i , hence a zero between each consecutive pair, as well as zeros at -1 and 1 . These are n distinct zeros, whereas the leading terms cancel and $\deg h \leq n - 1$, a contradiction. Thus one selected gap has $|q| \leq C_n$ throughout, and scaling back proves the bound.

For sharpness take the roots of q_* , namely

$$\frac{\cos((2k-1)\pi/(2n))}{r_n}, \quad 1 \leq k \leq n.$$

Their extreme roots are ± 1 , and every adjacent gap reaches C_n at a Chebyshev extremum. $\square$

Erdős, Herzog and Piranian already give a contained root segment in the collinear case [2, Theorem 1, p. 126]: if the zeros of a monic polynomial of degree n are real, lie in $[-1, 1]$ and have centroid in $[0, 1]$, then $\{|f| < 1\} \cap \mathbb{R}$ contains an interval holding at least $n/2$ of the zeros. Applied to q , after the reflection $w \mapsto -w$ if the centroid is negative, this gives for $n \geq 3$ two consecutive zeros joined by a segment of length at most D inside $\{|f| < (D/2)^n\}$. Theorem 14.1 adds the sharp level $C_n(D/2)^n$ and its equality configurations.

Critical sequences and the extremal configuration. For distinct real zeros, each gap between consecutive zeros contains exactly one zero of q' , and $|q|$ attains its maximum over the gap only there. The points c_i chosen in the proof are therefore the critical points of q , and the gap maxima are the moduli of its critical sequence $(q(c_1), \dots, q(c_{n-1}))$. A real polynomial with real zeros is determined by its critical sequence up to an increasing real affine change of variable [10, Theorem 1]; the proof above uses only the alternation count. In this language, the bound of Theorem 14.1 says that a real polynomial of degree n with n distinct real zeros, leading coefficient a and zero diameter D has a critical value of modulus at most $C_n |a| (D/2)^n$.

The comparison polynomial q_* is the monic polynomial of least deviation from zero on $[-1/r_n, 1/r_n]$, characterised by the equioscillation conditions recalled in [10, §1]. All $n - 1$ of its critical values have modulus C_n , so every critical point of q_* lies on the level curve $\{|q_*| = C_n\}$. Extremals with this property also occur in the level-curve length problem, where some extremal polynomial has all its critical points on $\{|p| = 1\}$ [9, Lemma 5], and in the sharp bound for $|f'|$ on a connected sublevel set $\{|f| \leq 1\}$ of a

monic polynomial, proved by Eremenko and Lempert [11, Theorem 1], whose equality cases are $e^{-in\theta}T_n(2^{1/n-1}e^{i\theta}z + b)$ with θ real [12, Theorem A].

If $D < 2$, the displayed level is below one and this segment lies in the open lemniscate. For $n \geq 3$, the same strict level holds already at $D = 2$; degree two is the exact boundary case. The proof is a complete authored ordinary argument. Two load-bearing alternation and Chebyshev-comparator kernels are Lean checked, while the affine normalization and transport remain ordinary; bounded replay is regression evidence only. This proves neither that every adjacent gap is safe nor any non-collinear analogue: the real order used in the alternation count is precisely what the general problem lacks.

15 A Poisson identity for critical-value means

Write FP_m for the equal-weight inequality $\sum_j (\prod_k |1 - \overline{c_k}c_j|)^{1/m} \leq m$.

The same points can define both an analytic geometric mean and a positive Poisson mixture. Their logarithmic-derivative relation converts the desired nonlinear estimate into a diagonal Taylor-coefficient inequality.

Theorem 15.1 (weighted free-point inequality in every degree). *Let $c_1, \dots, c_m \in \overline{\mathbb{D}}$ and let $w_j > 0$ satisfy $\sum_j w_j = 1$. Set*

$$G(z) = \prod_k |1 - \overline{c_k}z|^{w_k}.$$

Then

$$\sum_j w_j G(c_j)^2 \leq 1,$$

with equality if and only if every $c_j = 0$. Equal weights therefore give FP_m for every m .

Proof. First assume $|c_j| < 1$ and choose analytic logarithms zero at the origin. Put

$$g(z) = \exp \sum_j w_j \log(1 - \overline{c_j}z) = 1 + \sum_{\nu \geq 1} a_\nu z^\nu, \quad P = \sum_j w_j P_{c_j},$$

where $P_c(\zeta) = (1 - |c|^2)/|\zeta - c|^2$ and dm denotes normalised arclength. On the unit circle,

$$P = 1 - 2\Re \frac{\zeta g'}{g}.$$

Subharmonicity and Taylor orthogonality therefore give

$$\begin{aligned} \sum_j w_j G(c_j)^2 &\leq \int |g|^2 P \, dm \\ &= \int (|g|^2 - 2\Re(\zeta g' \overline{g})) \, dm \\ &= 1 - \sum_{\nu \geq 1} (2\nu - 1) |a_\nu|^2. \end{aligned}$$

All series converge on a neighbourhood of the closed disc in this case. For boundary points, replace c_j by rc_j and let $r \uparrow 1$. The finite left sum and every fixed Taylor coefficient converge; passing first with a finite coefficient sum proves the same inequality in

the limit. Equality forces $g = 1$. The rational function $\sum_j w_j \bar{c}_j / (1 - \bar{c}_j z)$ then vanishes identically. A nonzero point would give a pole with nonzero total positive weight, so all points are zero. Weighted Cauchy–Schwarz gives the equal-weight linear inequality. $\square$

The coefficient $a_1 = -\overline{\sum_j w_j c_j}$ gives the immediate refinement

$$\sum_j w_j G(c_j)^2 \leq 1 - \left| \sum_j w_j c_j \right|^2.$$

Thus the proof records a quantitative loss when the point centroid is nonzero. The specialised four-point and central-radius proofs are retained in the long record; the all-degree inequality above removes their cardinality restriction.

Relation to classical critical-value products. For a monic polynomial with all zeros in the unit disc, Dubinin [16, Theorem 2 and its proof, pp. 1172–1173] proves the sharp geometric-mean inequality $(\prod_{j=1}^{n-1} |f(c_j)|)^{1/(n-1)} \leq 1$. The proof uses the resultant identity (9), the maximum-modulus principle and Schur’s unit-circle Vandermonde inequality, not the dissymmetrisation argument of his Theorem 1. The positive-moment inequality below implies that product bound by AM–GM; the converse implication does not hold. This comparison identifies an antecedent, not a priority claim for the stronger moment statement.

Theorem 15.2 (critical-value mean in every degree). *Let f be monic of degree $n \geq 2$, with roots in a closed disc of radius R , and let $c_1, \dots, c_{n-1}$ be its critical points with multiplicity. Then*

$$\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1)R^{2n/(n-1)}.$$

In particular $(S)_n$, the inequality $\sum_j |f(c_j)|^{1/n} \leq (n-1)R$, holds in every degree.

Proof. Translation and scaling reduce to $R = 1$; the case $R = 0$ is immediate. Write $q = f'/n = \prod_j (z - c_j)$ and $q^\#(z) = \prod_j (1 - \bar{c}_j z)$. For roots strictly inside the unit disc, on $|\zeta| = 1$,

$$\Re \frac{\zeta f'(\zeta)}{f(\zeta)} = \sum_i \Re \frac{\zeta}{\zeta - z_i} > \frac{n}{2}.$$

Hence $|f - \zeta q| < |q| = |q^\#|$ there. Gauss–Lucas makes $q^\#$ zero-free on the closed disc, so the maximum principle gives $|f - \zeta q|/|q^\#| < 1$ inside. At a critical point,

$$|f(c_j)| \leq \prod_k |1 - \bar{c}_k c_j|.$$

The equal-weight quadratic theorem proves the required bound. Contracting a closed-disc root configuration and taking a limit treats its boundary. Power means give $(S)_n$, and scaling restores R . $\square$

If a zero is fixed at the disc centre, Dubinin's Theorem 3 [16, p. 1174] gives the sharper product estimate $(\prod_j |f(c_j)|)^{1/(n-1)} \leq (n-1)(|f'(0)|^2/n^n)^{1/(n-1)}$ in the unit-disc normalisation. Dubinin explicitly records the preceding contribution of Tischler [20, p. 444, as discussed by Dubinin]. Neither the product estimate nor its equality family settles the marked-zero positive-moment question in the research addendum [21].

The constant is attained by $f(z) = (z - h)^n - \lambda$ with $|\lambda| = R^n$. The proof above explains the analytic mechanism. The supplied formal index at commit 3d6d938d696f places the complete critical-value mean and the complete weighted free-point theorem in its checked build closure. The critical-value endpoint includes arbitrary centre, $R = 0$, and critical points counted with multiplicity. This is evidence recorded in the attached index, not a fresh kernel replay in this revision. Older candidate headers and narrower comparator selections do not supersede those complete targets. A critical-value mean supplies no attachment or path-length estimate by itself.

16 The exact nonlinear lifetime currency

The merge-tree route has its own exact theorem. For each nontrivial child component u , let k_u be its number of roots and let $r_u = \beta_u/\beta_{\text{parent}(u)} \in (0, 1]$ be its merge-scale ratio. Its contribution to the attachment age of each descendant root is

$$x_u = \frac{1}{k_u} \log \frac{1}{r_u}.$$

The edge lifetime is written

$$\lambda_k(q) = \log \frac{1 + q^{2/k}}{1 - q^{2/k}}, \quad I_k(r) = \int_r^1 \frac{dq}{q \lambda_k(q)}.$$

Theorem 16.1 (exact attachment-age/lifetime currency). *Define*

$$\Phi(x) = \int_0^x \frac{dt}{\log(\coth t)} \quad (x \geq 0).$$

At $t = 0$ the integrand is understood by its continuous limiting value 0 (equivalently, the integral is improper at that endpoint). For every integer $k \geq 1$ and $0 < r \leq 1$, with $x = k^{-1} \log(1/r)$,

$$I_k(r) = k\Phi(x). \tag{O1}$$

The function Φ is increasing and strictly convex on $(0, \infty)$, and

$$\frac{\Phi(x)}{x} \rightarrow 0 \quad (x \downarrow 0). \tag{O2}$$

Consequently, for every fixed $k \geq 1$ and every $c > 0$, some $0 < r < 1$ satisfies

$$I_k(r) < c \frac{1}{k} \log \frac{1}{r}. \tag{O3}$$

In particular, there is no positive universal linear conversion from attachment age to lifetime.

Proof. Set $q = e^{-kt}$. Then $dq/q = -k dt$, the endpoints $q = 1, r$ become $t = 0, x$, and

$$\frac{1 + e^{-2t}}{1 - e^{-2t}} = \coth t,$$

which proves (O1). Since $\coth t$, and hence $\log(\coth t)$, is strictly decreasing, the positive integrand $1/\log(\coth t)$ is strictly increasing. This proves monotonicity and strict convexity. Moreover

$$0 \leq \frac{\Phi(x)}{x} \leq \frac{1}{\log(\coth x)} \rightarrow 0,$$

which gives (O2); taking $r = e^{-kx}$ gives (O3). $\square$

The identity explains why a linear charge per attachment age loses the behaviour of very young edges. Chain allocation must retain the distribution of ages and the convex function Φ . The weighted Jensen bounds and the two-young-root selector are retained in [attachment age lifetime orlicz](#). They still require a geometric compatibility estimate before they produce a contained connector.

17 The remaining geometric selection problem

Let $\mathcal{K}_n$ be the compact coefficient class of monic degree- n polynomials with roots in the closed unit disc. Define

$$\Lambda(f) = \inf_{\gamma} \text{length}(\gamma),$$

where $\gamma \subset \{|f| \leq 1\}$ joins two different listed root occurrences. A repeated root permits a constant curve.

A dense generic class suffices. The functional Λ is lower semicontinuous. Indeed, bounded-length competitors have uniformly bounded images and admit constant-speed parametrisations with uniformly bounded Lipschitz constants. Arzelà–Ascoli, convergence of root multisets, and uniform convergence of the polynomials on compact sets produce a limiting competitor. Its length is no larger than the lower limit of the approximating lengths, and closed sublevel containment passes to the limit. Thus a dense generic bound $\Lambda \leq 2$ proves that bound on all of $\mathcal{K}_n$. For an open-disc polynomial, enclosing its roots in a disc of radius $R < 1$ and scaling gives a path with length at most $2R < 2$ in $\{|f| \leq R^n\} \subset \{|f| < 1\}$.

For $f(z) = z^n - 1$, the punctured unit sublevel has separate root sectors. Every path between distinct roots passes through zero, and the two radial segments attain length 2. The exact closed-class question is therefore

$$\boxed{\Lambda(f) \leq \Lambda(z^n - 1) = 2 \quad (f \in \mathcal{K}_n).}$$

The lower-semicontinuity argument is proved in [generic sufficiency closure](#).

A sufficient canonical-arc estimate. On the generic simple-critical, ray-separated class, cutting the value plane along critical rays gives the finite inverse-sheet construction recorded in [attachment aware reeb](#). The underlying inverse-sheet construction is classical: Dubinin [16, §1, pp. 1169–1170] cuts outward from the critical values and describes the resulting sheet adjacency tree; [22, §2, pp. 34–35] uses a related network in a capacity argument. The latter assumes critical values bounded by one and controls logarithmic capacity, not Euclidean length. Here the task is to specify admissible descent arcs and estimate their metric cost. Let $L_f(c)$ be the length of the canonical two-root descent arc through a critical point c ; its two halves are Newton-flow trajectories from c to roots, that is, edges of the Newtonian graph [6, Definition 2.1]. Only critical points with $|f(c)| \leq 1$ give admissible unit-level candidates. The sufficient metric target is

$$\min_{\substack{f'(c)=0 \\ |f(c)| \leq 1}} L_f(c) \leq 2.$$

Since Λ minimises over all curves, this canonical minimum is an upper bound for Λ . Equality between the two minimisations is not assumed. The ordinary slit-sheet construction closes its stated generic topology problem; formalising that construction remains a separate task.

The attachment obstruction is quantitative. For $f(z) = z^n - r^n$, the inverse image of the positive value ray with argument θ approaches the saddle at zero only if its value-plane direction approaches π . A lift meets $B(0, \varepsilon)$, with $0 < \varepsilon < r$, precisely when

$$\cos \theta < 0, \quad |\sin \theta| < (\varepsilon/r)^n.$$

The total angular window is $2 \arcsin((\varepsilon/r)^n)$. An angular mean length bound therefore gives no uniform control at the shrinking set of attachable directions. A successful fixed-attachment estimate needs this angular concentration or a construction free to choose its attachment. The binomial itself is solved by radial segments.

Which information is insufficient. The sextic of Proposition 1.2 excludes automatic control of every prescribed root spoke once further partial sums occur. The exact Cassini example in Section 11 excludes the historical global strip-tree budget even after a local saddle model is corrected. The examples in Section 7 show that the least critical modulus and a separation-two test do not cover all polynomials, and that initial merge arity does not control capacity after subsequent mergers. The all-degree critical-value inequality controls a marginal distribution; it does not identify compatible inverse branches. The ordinary note [blaschke power critical spectra](#) records Blaschke-power polynomials $F_N = A^N - D^N$ that realise the sharp critical-value moment numerator $4s$ inside the polynomial class, including a high-critical nonseparated regime with exact rational certificates. The same family has ordinary short connectors of length $O(1/N)$ and an exact degree-dependent degeneration of Λ . These statements are not parent proofs, are not Lean-checked as analytic theorems, and do not replace the trinomial flagship.

The previous return also supplies a near-regular-pentagon family for which *all* origin spokes fail [21, Proposition A of the preserved earlier note]. The accompanying

addendum analyses every critical point of that family and gives an ordinary counterargument to a critical *straight*-hub selector even locally near the regular pentagon. That new argument is separated from the paper's registered formal results and requires independent review. It does not exclude a noncritical vertex, curved inverse-ray arms or an arbitrary contained connector.

The remaining parent regime has $\mu > 13/25$ and lies outside the solved families and the applicable separation conditions. Further progress requires a contained pair selected with its geometric cost, a quantitative attachment estimate, or a different competitor for Λ . The failed scalar implications and their exact witnesses are preserved in the long record.

18 Proof boundaries

The evidence belongs to each statement, not to a blanket label for the whole paper. The supplied index at commit 3d6d938d696f records complete checked targets for the critical-value mean, weighted free-point inequality and translated cubic-fibre path. The trinomial radial bounds and radius sum, together with the named finite selectors and algebraic kernels, have their own cited formal statements. A comparator is a selected interface, not an inventory of every formal theorem.

The complete degree-three and primitive-quintic path declarations also occur in the supplied source, but are indexed outside the checked build. Their ordinary proofs remain mathematical arguments; absence of an attached build receipt is not an absence of an ordinary proof or a formal-source file.

The $13/25$ theorem uses ordinary analysis and an exact rational certificate; its scaled corollary, the independent $71/10$ argument, inverse-map geometry and slit-sheet topology are not certified merely by the neighbouring finite Lean kernels. The new research addendum is ordinary, unreviewed work, with its rational sign checks explicitly separated from its analytic steps. No fresh Lean replay or independent human review is claimed in this revision.

Fixed-degree lower semicontinuity reduces the closed class to a dense generic metric bound. The admissible *curved* canonical-arc estimate is a sufficient target, not an identity for the unrestricted path infimum. Raster searches are finite-sample computations, not proofs of that target.

Sources and adjacent results

The original root-to-root problem is [2, Problem 5, p. 139]. Pendyala's degree-four theorem [4, Theorem 1] is directly comparable. Erdős Problem #1120, treated by Pendyala [5, Definition 1.1 and Theorem 1.2], asks for a path from the origin to the unit circle inside a filled sublevel set; its endpoints differ, and it does not imply the root-pair conclusion. The exact Cassini obstruction in Section 11 concerns the metric assertion of [3, Prop. 12]. Eremenko and Hayman prove the global boundary estimate $\text{length}\{z : |p(z)| = 1\} < 9.173 \deg p$ and show that an extremal level set is connected [9, Thm. 1 and the lemma on connectedness]. Their metric is the total length of a level curve, so this does not supply the short internal root-to-root connector asked for here.

The Newton value identity of Section 8 is a lemma of Shub, Tischler and Williams, and the ray-segment description of Newton-flow edges belongs to their Newtonian graph; we use both in the form recorded by Kozen and Stefánsson [6, Lemmas 2.1–2.2 and §2]. The area arguments use Pólya’s inequality and the area–capacity inequality in the formulations [7, Theorems 1 and 6]; the square-root uniformisation uses the component-wise Riemann–Hurwitz count in the proof of [8, Proposition 2.1]. Existence of a contained root segment for collinear zeros already follows from [2, Theorem 1]; Section 14 adds the sharp constant. Crane’s Smale mean-value argument [17, Lemma 2.1, Theorem 2.2 and §§3–4] combines inverse branches, radial slits and capacity, but concerns a derivative-normalised mean-value ratio rather than our absolute monic critical values. Dubinin’s four-point theorem [18, Theorem 1 and Corollary 4] assumes a bound on *all* critical values and concerns distortion; its normalisation does not supply our root-disc moment or path estimate. Kuznetsova and Tkachev [19, Theorems 1–2] study length functions on regular lemniscate levels. Their log-convexity is useful context, not a uniform one-root perimeter bound at a critical endpoint. Downloaded-source digests, precise locators, and withdrawal history are retained in the long record. No uniqueness or priority claim is inferred from this comparison.

A Guide to the formal sources

The attached declaration index, not a single Newton-flow module or a shared historical macro, is the source map for this revision. At snapshot 3d6d938d696f, the following distinctions are essential.

- Complete targets in the supplied checked closure: [critical-value mean](#), [weighted free-point inequality](#), and [translated cubic-fibre path](#).
- Complete implementations indexed outside that closure: [degree-three path](#).
- Admitted to the checked ErdosProblems root in this revision: [primitive-quintic path](#).
- Finite selected kernels, source-only release declarations and ordinary analytic assemblies have separate rows in the accompanying formal-scope map. The older Newton declaration list is preserved verbatim in the editorial transfer archive; it remains useful for those lemmas, not as complete paper coverage.

No source-link reachability check or fresh kernel build is implied by these snapshot links. The original paper-wide commit macro is retained for legacy links; newer target links name their own snapshot explicitly.

B Pinned Lean sources

The following declarations are this note’s pin-faithful source inventory. Line numbers are those of the commit named by `\commit`.

- [newton flow vector](#)
- [derivative mul newton flow vector](#)

- same positive ray
- same positive ray refl
- same positive ray symm
- same positive ray trans
- real affine line
- real affine line eq affine span
- is closed real affine line
- dense compl real affine line
- exists small avoiding finite real affine lines
- same positive ray imp mem real affine line
- norm lt one of near root
- constant perturbation root near original
- same positive ray of real exp decay

References

- [1] T. F. Bloom, *Erdős Problems*, problem 1041. <https://www.erdosproblems.com/1041>
- [2] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148, doi:[10.1007/BF02790232](https://doi.org/10.1007/BF02790232).
- [3] shtuka, *A Short Path Joining Two Zeros Inside a Polynomial Lemniscate*, manuscript posted 24 March 2026, 48 pp. <https://shtuka123.github.io/1041/main.pdf>
- [4] V. S. Pendyala, *A Degree-Four Lemniscate Path Theorem*, arXiv:[2606.24875v1](https://arxiv.org/abs/2606.24875) (2026), doi:[10.48550/arXiv.2606.24875](https://doi.org/10.48550/arXiv.2606.24875).
- [5] V. S. Pendyala, *Shortest paths in polynomial lemniscate sublevel sets and a problem of Erdős*, arXiv:[2606.19178v1](https://arxiv.org/abs/2606.19178) (2026), doi:[10.48550/arXiv.2606.19178](https://doi.org/10.48550/arXiv.2606.19178).
- [6] D. Kozen and K. Stefánsson, *Computing the Newtonian graph*, J. Symbolic Comput. **24** (1997), no. 2, 125–136, doi:[10.1006/jSCO.1997.0118](https://doi.org/10.1006/jSCO.1997.0118); authors' copy <https://www.cs.cornell.edu/kozen/Papers/newton.pdf>.
- [7] E. Crane, *The areas of polynomial images and pre-images*, Bull. London Math. Soc. **36** (2004), no. 6, 786–792, doi:[10.1112/S0024609304003509](https://doi.org/10.1112/S0024609304003509); preprint arXiv:[math/0302189v1](https://arxiv.org/abs/math/0302189), whose statement numbers are cited.
- [8] P. Ebenfelt, D. Khavinson, and H. S. Shapiro, *Two-dimensional shapes and lemniscates*, in *Complex Analysis and Dynamical Systems IV, Part 1*, Contemp. Math. **553**, Amer. Math. Soc., Providence, RI, 2011, 45–59, doi:[10.1090/conm/553/10931](https://doi.org/10.1090/conm/553/10931); preprint arXiv:[1003.4567v1](https://arxiv.org/abs/1003.4567), whose statement numbers are cited.

- [9] A. Eremenko and W. Hayman, *On the length of lemniscates*, Michigan Math. J. **46** (1999), no. 2, 409–415, doi:[10.1307/mmj/1030132418](https://doi.org/10.1307/mmj/1030132418); preprint arXiv:[0805.2295](https://arxiv.org/abs/0805.2295).
- [10] A. Eremenko and P. Yuditskii, *Comb functions*, Contemp. Math. **578** (2012), 99–118, doi:[10.1090/conm/578/11472](https://doi.org/10.1090/conm/578/11472); preprint arXiv:[1109.1464v1](https://arxiv.org/abs/1109.1464v1).
- [11] A. Eremenko and L. Lempert, *An extremal problem for polynomials*, Proc. Amer. Math. Soc. **122** (1994), no. 1, 191–193, doi:[10.1090/S0002-9939-1994-1207536-1](https://doi.org/10.1090/S0002-9939-1994-1207536-1).
- [12] A. Eremenko, *A Markov-type inequality for arbitrary plane continua*, Proc. Amer. Math. Soc. **135** (2007), no. 5, 1505–1510, doi:[10.1090/S0002-9939-06-08640-0](https://doi.org/10.1090/S0002-9939-06-08640-0); preprint arXiv:[math/0606745v1](https://arxiv.org/abs/math/0606745v1).
- [13] C. J. Bishop, A. Eremenko, and K. Lazebnik, *On the shapes of rational lemniscates*, Geom. Funct. Anal. **35** (2025), no. 2, 359–407, doi:[10.1007/s00039-025-00704-2](https://doi.org/10.1007/s00039-025-00704-2); preprint arXiv:[2407.14610v1](https://arxiv.org/abs/2407.14610v1).
- [14] V. N. Dubinin, *Some inequalities for polynomials and rational functions associated with lemniscates*, Zap. Nauchn. Sem. POMI **404** (2012), 83–99; English translation, J. Math. Sci. **193** (2013), no. 1, 45–54, doi:[10.1007/s10958-013-1432-4](https://doi.org/10.1007/s10958-013-1432-4).
- [15] G. Pólya, *Beitrag zur Verallgemeinerung des Verzerrungssatzes auf mehrfach zusammenhängende Gebiete*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse (1928), printed pp. 228–232 and 280–282. <https://archive.org/details/sitzungsbericht1928preu>.
- [16] V. N. Dubinin, *Inequalities for critical values of polynomials*, Sb. Math. **197** (2006), no. 8, 1167–1176, doi:[10.1070/SM2006v197n08ABEH003793](https://doi.org/10.1070/SM2006v197n08ABEH003793).
- [17] E. Crane, *A bound for Smale’s mean value conjecture for complex polynomials*, Bull. London Math. Soc. **39** (2007), no. 5, 781–791, doi:[10.1112/blms/bdm063](https://doi.org/10.1112/blms/bdm063); author preprint <https://people.maths.bris.ac.uk/~maetc/SMVCbound.pdf>.
- [18] V. N. Dubinin, *Four-point distortion theorem for complex polynomials*, arXiv:[1301.3985v1](https://arxiv.org/abs/1301.3985v1) (2013).
- [19] O. S. Kuznetsova and V. G. Tkachev, *Length functions of lemniscates*, Manuscripta Math. **112** (2003), 519–538, doi:[10.1007/s00229-003-0411-3](https://doi.org/10.1007/s00229-003-0411-3); preprint arXiv:[math/0306327](https://arxiv.org/abs/math/0306327).
- [20] D. Tischler, *Critical points and values of complex polynomials*, J. Complexity **5** (1989), no. 4, 438–456.
- [21] W. Cook / Plectis, *Three refinements for the lemniscate-path programme*, 16 September 2026, accompanying unreviewed research note, Sections 1–3, with the earlier *Structural obstructions* note preserved in the same revision bundle.
- [22] V. N. Dubinin, *Lemniscates and inequalities for the logarithmic capacities of continua*, Mat. Zametki **80** (2006), no. 1, 33–37; English translation, Math. Notes **80** (2006), no. 1, 31–35, doi:[10.1007/s11006-006-0105-8](https://doi.org/10.1007/s11006-006-0105-8).

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.